

(1001CJA102120128)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

FULL SYLLABUS

28-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	D	D	C	A,C,D	A,B,C,D	A,C	B,C	A,B,C,D
SECTION-II	Q.	11	12	13	4	5					
	A.	B,C	A,D	A,B,C							
SECTION-II	Q.	1	2	3	4	5					
	A.	331	231	211	213	111					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A or C	A	B	A	A,C	A,C or A	A,D	A,B,C,D	B,D
SECTION-II	Q.	11	12	13	4	5					
	A.	B,C,D	B,C,D	A,C							
SECTION-II	Q.	1	2	3	4	5					
	A.	231	312	233	312	213					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	B	A	B	A,B,C	B,C	D	A,D	A,B,D
SECTION-II	Q.	11	12	13	4	5					
	A.	A,C	B,C,D	B,C,D							
SECTION-II	Q.	1	2	3	4	5					
	A.	111	123	132	123	132					

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SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	A	C	D	D	A,B	A,C	A,B	B,C
SECTION-II	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C	A,C	A,B,C	B,C	A	B	A	C		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	B	B	C	D	A,C,D	A,B	A,B,C,D	B,D
SECTION-II	Q.	11	12	13	14	15	16	17	18		
	A.	A,B,C,D	B,C,D or C,D	A,B,C,D	A,B,C	D	B	A	B		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	B	B	A	C	A,D	B,D	B,C	B,D
SECTION-II	Q.	11	12	13	14	15	16	17	18		
	A.	B,C	C,D	A,B	B,C,D	A	B	B	A		



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PAPER-1

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (C)

Sol. REFER HCV PART-I, PAGE NO. 244

2. Ans. (A)

$$\text{Sol. } v_{p \max} = A\omega \sin\left(\frac{\pi}{6}\right)$$

$$\omega = 2\pi f = 2\pi \frac{v}{\lambda} \text{ \& } \lambda = 0.4 \text{ m}$$

3. Ans. (D)

$$\text{Sol. } I = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \phi,$$

$$\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} \frac{\lambda}{4} = \frac{\pi}{2}$$

$$\text{so } I = 2I_0$$

4. Ans. (D)

$$\text{Sol. } d \sin \theta = n\lambda$$

$$d = \frac{n\lambda}{\sin \theta} = \frac{\lambda}{\sin \theta} = \frac{0.12}{3} \times 5$$

$$d = \frac{0.6}{3} = 0.2 \text{ m}$$

5. Ans. (C)

Sol. Lenses are of opposite nature, so,

$$\text{in water, } \frac{1}{f_1} = \left(\frac{\mu_1}{\mu_w} - 1\right) \left(\frac{2}{R}\right)$$

$$\frac{1}{f_2} = \left(\frac{\mu_2}{\mu_w} - 1\right) \left(-\frac{2}{R}\right)$$

focal length of the system is

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$\frac{1}{24} = \left(\frac{\mu_1}{\mu_w} - \frac{\mu_2}{\mu_w}\right) \left(\frac{2}{R}\right)$$

$$= \left(\frac{\mu_1 - \mu_2}{\mu_w}\right) \left(\frac{2}{20}\right)$$

$$\text{So, } (\mu_1 - \mu_2) = \frac{5}{9}$$

6. Ans. (A, C, D)

$$\text{Sol. } T_A = T_B = (300 + 400)/2 = 350^\circ \text{C}$$

$$\therefore T_A = T_B = T_C$$

7. Ans. (A, B, C, D)

$$\text{Sol. } Ft = mv$$

$$W = \frac{1}{2}mv^2 = \frac{1}{2}mF^2t^2$$

$$W = \frac{F^2t^2}{2m}$$

8. Ans. (A, C)

9. Ans. (B, C)

10. Ans. (A, B, C, D)

11. Ans. (B, C)

$$\text{Sol. } \text{Largest face} = 10.4 \times 3.8 = 39.52$$

ans will have 2 sig figure

Hence ≈ 40 is correct

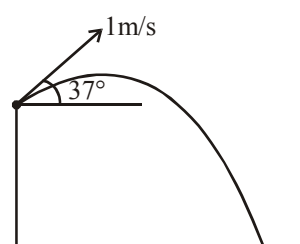
$$\rightarrow \text{Smallest face} = 3.8 \times 1.02 = 3.876$$

ans will have 2 sig figure

Hence $\approx 3.9 \text{ cm}^2$

$$\rightarrow \text{Volume} \approx 40 = 4.0 \times 10^1 \text{ cm}^3$$

12. Ans. (A, D)



Sol.

$$v = \sqrt{2gh} = \sqrt{2 \times 10 \times \frac{5}{100}} = 1 \text{ m/s}$$

$$\frac{-8}{100} = \frac{3}{5}t - \frac{1}{2} \times 10 \times t^2 \Rightarrow t = \frac{1}{5} \text{ sec}, \frac{-8}{100} \text{ sec}$$

$$\therefore x = \frac{4}{5} \times \frac{1}{5} \times 100 = 16 \text{ cm}$$

∴ horizontal range from point B = 16 – 6
= 10 cm

Force required to keep the vessel in static equilibrium.

$$= \rho A v^2 \cos 37$$

$$= 1000 \times \frac{3}{2} \times 10^{-4} \times 1 \times \frac{4}{5} = 0.12 \text{ N}$$

13. Ans. (A,B,C)

Sol. $^{55}\text{Mn} + {}^1_1\text{H} \rightarrow {}^{56}\text{Mn} + {}^1_1\text{H}$

$$\text{Steady state} \Rightarrow \frac{dN}{dt} = \frac{dN}{dt}$$

$$\Rightarrow \frac{dN}{dt} = \lambda N_0 = 6.93 \times 10^{10}$$

$$= \frac{0.693}{3 \times 3600} N_0 = 6.93 \times 10^{10}$$

$$N_0 = 10800 \times 10^{11}$$

$$m = \frac{N_0}{N_a} \times 56 \text{ gm}$$

$$= \frac{10800 \times 10^{11} \times 56}{6 \times 10^{23}} \approx 100 \text{ ng}$$

SECTION-II

1. Ans. 331

Sol. (A) For system A+B, $(4+6) v_0 = 4 \times 5 + 6 \times 10$
 $\Rightarrow v_0 = 8 \text{ m/s}$

$$(B) v_{\text{CM}} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{4 \times 5 + 6 \times 10}{4 + 6} = 8 \text{ m/s}$$

(C) Work done by friction on

$$B = \frac{1}{2} \times 6 \times 8^2 - \frac{1}{2} \times 6 \times 10^2 = -108 \text{ J}$$

2. Ans. 231

3. Ans. 211

4. Ans. 213

Sol. (i) thermal resistance = $i = \frac{\theta}{t} = \frac{\Delta T}{R_t}$

$$= \frac{ML^2 T^{-2}}{T \theta} = \frac{1}{R_t}$$

$$R_t = M^{-1} L^{-2} T^3 \theta$$

$$(ii) \frac{\theta}{t} = \sigma A T^4$$

$$\sigma = \frac{\theta}{t A T^4} = M T^{-3} \theta^{-4}$$

$$(iii) \lambda T = b(mk)$$

$$\lambda = \frac{b}{T} = L \theta$$

$$(iv) \text{Heat current} = \frac{\theta}{t} = \frac{ML^2 T^{-2}}{T}$$

$$i = \frac{\theta}{t} = ML^2 T^{-3}$$

5. Ans. 111

Sol. (A) $Q = Q_0 e^{-t/RC}$

$$i = \frac{dQ}{dt} = \frac{-Q_0}{RC} e^{-t/RC} = i_0 e^{-t/RC}$$

$$(B) i = \frac{V}{R} e^{-t/RC}$$

$$i = \frac{V}{R} (1 - e^{-Rt/L})$$

$$(D) i = i_0 e^{-Rt/L}$$

PART-2 : CHEMISTRY

SOLUTION

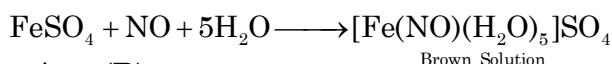
SECTION-I

1. Ans.(D)

2. Ans.(A or C)

3. Ans.(A)

Sol.



Brown Solution

4. Ans.(B)

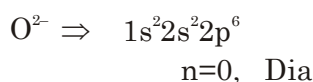
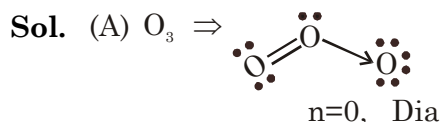
5. Ans.(A)

6. Ans.(A,C)

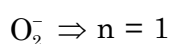
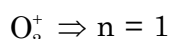
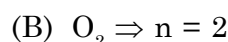
7. Ans.(A,C or A)

8. Ans.(A,D)

9. Ans.(A,B,C,D)



n=0, Dia



5. Ans. (B)

Sol. Let $z = e^{i\theta}$

$$\begin{aligned} \text{So, } w &= \frac{e^{i\theta} - 1}{e^{i\theta} + 1} = \frac{(e^{i\theta} - 1)(e^{-i\theta} + 1)}{|e^{i\theta} + 1|^2} \\ &= \frac{e^{i\theta} - e^{-i\theta}}{2 + 2\cos\theta} = \frac{2i\sin\theta}{2(1 + \cos\theta)} = i \tan\left(\frac{\theta}{2}\right) \end{aligned}$$

6. Ans. (A,B,C)

Sol. $(\sin^{-1}x)^2 - (\cos^{-1}x)^2 = \frac{\pi}{2}(\sin^{-1}x - \cos^{-1}x)$

$$= \frac{\pi}{2}\left(\frac{\pi}{2} - 2\cos^{-1}x\right) \in \left[-\frac{3\pi^2}{4}, \frac{\pi^2}{4}\right] \subset (-8, 3)$$

Also, $\sin^{-1}x + 2\cos^{-1}(-x) = 2\pi$

$$\Rightarrow \sin^{-1}x + 2(\pi - \cos^{-1}x) = 2\pi$$

$$\Rightarrow \sin^{-1}x - 2\left(\frac{\pi}{2} - \sin^{-1}x\right) = 0$$

$$\Rightarrow \sin^{-1}x = \frac{\pi}{3} \Rightarrow x = \frac{\sqrt{3}}{2} > \frac{1}{2}$$

7. Ans. (B,C)

Sol. $A^2 = I \Rightarrow (\det A)^2 = \det(I)$

$$\Rightarrow \det A = \pm 1$$

$$\text{so, } \det((A - I)^3 + (A + I)^3 - 7A)$$

$$= \det(2A^3 + 6A - 7A) = \det(2A^2A - A)$$

$$= \det(A) = \pm 1$$

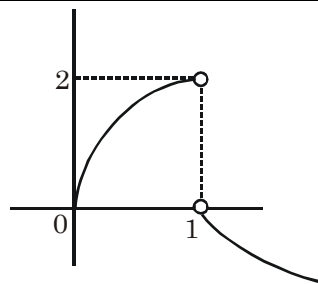
8. Ans. (D)

Sol. Function $|t|$ is not differentiable only at $t = 0$, whereas function $|\sin\pi t|$ is differentiable everywhere except at all integral points. Also, function $\sqrt[3]{t-2}$ is not differentiable only at $t = 2$. So, $\vec{r}(t)$ is differentiable everywhere except at all integral points.

9. Ans. (A,D)

Sol. At $x = 1$, $f(x)$ has local

$$\begin{cases} \text{maxima ; if } p \geq 2 \\ \text{minima ; if } p < 0 \end{cases}$$



10. Ans. (A,B,D)

Sol. $\int e^x (x+1)^2 dx = \int e^x ((x^2+1) + 2x) dx$
 $= e^x(x^2+1) + c$
 (as $\int e^x (f + f') dx = e^x f + c$)

11. Ans. (A,C)

Sol. Given identity

$$\begin{aligned} \int_0^\infty \frac{x^2}{(x^2+a^2)(x^2+b^2)(x^2+c^2)} dx \\ = \frac{\pi}{2(a+b)(b+c)(c+a)} \end{aligned}$$

For I, put $a = 0$, $b = 2$ and $c = 3$:

$$\int_0^\infty \frac{dx}{(x^2+4)(x^2+9)} = \frac{\pi}{60} = I \quad \dots(i)$$

Now, put $a = 1$, $b = 2$ and $c = 3$:

$$\int_0^\infty \frac{x^2 dx}{(x^2+1)(x^2+4)(x^2+9)} = \frac{\pi}{120} \quad \dots(ii)$$

so, (i) - (ii) :

$$\int_0^\infty \frac{dx}{(x^2+1)(x^2+4)(x^2+9)} = \frac{\pi}{120} = J$$

12. Ans. (B,C,D)

Sol. $\Delta = \begin{vmatrix} \sin\theta & \cos\theta & \cos\theta \\ \cos\theta & \sin\theta & \cos\theta \\ \cos\theta & \cos\theta & \sin\theta \end{vmatrix}$

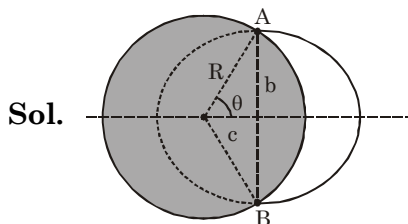
$$= (\sin\theta + 2\cos\theta) \begin{vmatrix} 1 & 1 & 1 \\ \cos\theta & \sin\theta & \cos\theta \\ \cos\theta & \cos\theta & \sin\theta \end{vmatrix}$$

$$[R_1 \rightarrow R_1 + R_2 + R_3]$$

$$= (\sin\theta + 2\cos\theta) \begin{vmatrix} 1 & 0 & 0 \\ \cos\theta & \sin\theta - \cos\theta & 0 \\ \cos\theta & 0 & \sin\theta - \cos\theta \end{vmatrix}$$

$$\begin{bmatrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{bmatrix}$$

$$= (\sin\theta - \cos\theta)^2 (\sin\theta + 2\cos\theta)$$

PAPER-2**PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (B)**

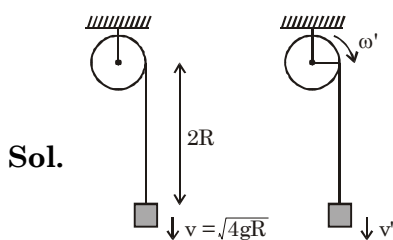
$$\frac{A_1}{t_1} = \frac{A_2}{t_2}$$

$$\frac{t_2}{t_1} = \frac{A_2}{t_1} = \frac{\frac{\pi ab}{2} + \frac{1}{2}(2b)(c)}{\pi ab}$$

$$\frac{t_2}{t_1} = \frac{1}{2} + \frac{c}{a\pi} = \frac{1}{2} + \frac{\cos \theta}{\pi} = \frac{1}{2} + \frac{1}{2\pi} = \frac{1}{2} \left(1 + \frac{1}{\pi} \right)$$

$$\approx \frac{1}{2} \left(1 + \frac{1}{3} \right)$$

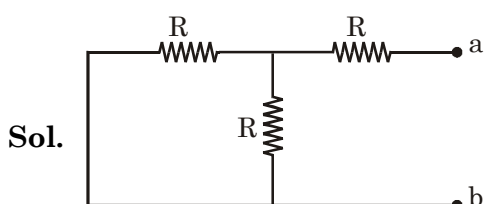
$$\frac{t_2}{t_1} \approx \frac{2}{3}$$

2. **Ans. (A)**

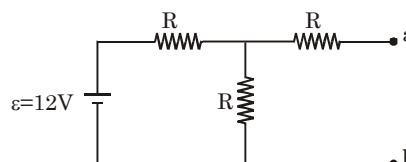
$$-J = mv' - mv$$

$$JR = I_C \omega'$$

$$v' = \omega' R$$

3. **Ans. (A)**

$$R_{ab} = \frac{R}{2} + R = 3\Omega$$



$$V_{ab} = \frac{\epsilon}{2R} (R) = \frac{\epsilon}{2} = 6V$$

$$P_{\max} = \frac{(V_{ab})^2}{4(R_{ab})} = \frac{(6)^2}{4 \times 3} = \frac{36}{12}$$

$$P_{\max} = 3 \text{ Watt}$$

4. **Ans. (C)**

Sol. $\Delta U = nC_v \Delta T$

$$= n \frac{3}{2} R \Delta T = \frac{3}{2} (nR \Delta T)$$

$$\Delta U = \frac{3}{2} [P_f V_f - P_i V_i]$$

$$= \frac{3}{2} [3 \times 10^5 \times 3 \times 10^{-3} - 3 \times 10^5 \times 1 \times 10^{-3}]$$

$$= \frac{3}{2} \times 3 \times 10^{-3} \times 10^5 [1]$$

$$X = \Delta U = 4.5 \times 10^2 \text{ Joule}$$

5. **Ans. (D)**

Sol. $KE_e = (41.8 - 13.6) \text{ eV}$

$$\frac{1}{2} mv^2 = KE_e = 28.2 \text{ eV}$$

6. **Ans. (D)**

Sol. Mixture has activity of 1000 Bq.

After 1hr, activity is 900 Bq

So, activity of A in initial mixture = 200 Bq

Activity of B initially in mixture = 800 Bq

$$\text{Activity after 20 years} = (800) \left(\frac{1}{2} \right)^2$$

$$= 200 \text{ Bq}$$

Assuming substance A completely current electricity

7. Ans. (A,B)

Sol. $T = \frac{2u}{g_{\text{eff}}} = \frac{2 \times 8}{12} = \frac{4}{3} \text{ s}$

$$h = \frac{u^2}{2g_{\text{eff}}} = \frac{8^2}{24} = \frac{8}{3} \text{ m}$$

$$h = \frac{u^2}{2g_{\text{eff}}} = \frac{8^2}{24} = \frac{8}{3} \text{ m}$$

8. Ans. (A,C)

Sol. $\frac{T_s}{a_s} = \frac{T_b}{a_b} \dots (i)$

$$T_s(2-x) = T_b \cdot x \dots (ii)$$

From (i) & (ii), $x = 2 \times \frac{0.1}{0.3} = \frac{2}{3} \text{ m}$

$$\frac{T_s}{a_s Y_s} = \frac{T_b}{a_b Y_b} \dots (iii)$$

$$T_s(2-y) = T_b y \dots (iv)$$

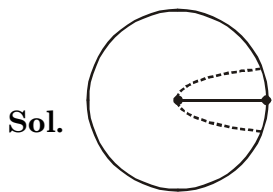
From (iii) & (iv) $y = 1 \text{ m}$

9. Ans. (A,B)

10. Ans. (B,C)

11. Ans. (A,B,C)

12. Ans. (A,C)



$$R = \frac{\lambda}{4} \Rightarrow \lambda = 4R$$

$$f = \frac{v}{\lambda} = \frac{1}{4R} \sqrt{\frac{B}{\rho}}$$

$$A_x = A \sin kx = A \sin \left(\frac{2\pi}{4R} \times \frac{R}{2} \right) = \frac{A}{\sqrt{2}}$$

$$u_{\text{max}} = A\omega$$

$$= \frac{\pi A}{2R} \sqrt{\frac{B}{\rho}}$$

13. Ans. (A,B,C)

Sol. $2 \frac{di}{dt} = 10 - 2i$

$$\frac{di}{dt} + i = 5$$

$$\therefore i = 5(1 - e^{-t})$$

$$\text{at } t = \ln 2, i = 5 \left(1 - \frac{1}{2} \right) = 2.5 \text{ A}$$

$$\therefore U = \frac{1}{2} Li^2 = 6.25 \text{ J}$$

$$\frac{dU}{dt} = \frac{1}{2} \cdot L \cdot 2i \frac{di}{dt} = Li \frac{di}{dt} = 2 \times 2.5 \times 2.5 = 12.5 \text{ J/s}$$

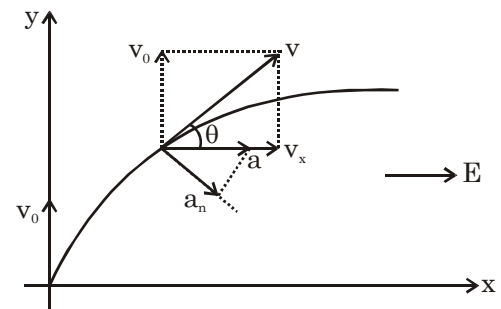
14. Ans. (B,C)

Sol. $v^2 = v_0^2 + 2 \left(\frac{qE_0}{m} \right) x_0$

$$v = \sqrt{2} v_0$$

$$a_n = a \sin \theta$$

$$a_n = \frac{qE_0}{m} \left(\frac{v_0}{v} \right) = \frac{qE_0}{m\sqrt{2}}$$



$$R = \frac{v^2}{a_n} = \frac{(\sqrt{2}v_0)^2 m\sqrt{2}}{qE_0}$$

$$R = \left(\frac{mv_0^2}{qE_0} \right) (2\sqrt{2})$$

$$R = 4\sqrt{2}x_0$$

15. Ans. (A)

Sol. $V_A - V_C = V_A - V_B = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{r_B}{a}$

$$= \frac{6 \times 10^{-7} \times 2}{4\pi\epsilon_0} \ln \frac{7.5}{2.5}$$

$$= 12 \times 10^{-7} \times 9 \times 10^9 \ln 3$$

$$= 118.65 \times 10^2$$

$$= 11.865 \times 10^3$$

$$\approx 12 \text{ kV}$$

16. Ans. (B)

Sol. $\Delta K = e(V_A - V_B)$

$$= 1.6 \times 10^{-19} \times 12 \times 10^{-3}$$

$$= 19.2 \times 10^{-16} = 1.92 \times 10^{-15}$$

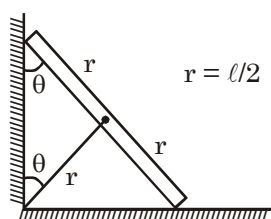
$$\approx 2 \times 10^{15} \text{ J}$$

17. Ans. (A)

18. Ans. (C)

Sol. Q. No. 17 & 18

When ladder is in contact with wall COM moves in a circle of radius r . The horizontal component of velocity of centre of mass will increase until the rod is in contact with vertical wall. When contact is lost it become constant in the absence of any horizontal force.



Conservation of mechanical energy gives,

$$mg \frac{1}{2} (1 - \cos \theta) = \frac{1}{2} MV_c^2 + \frac{1}{2} I_c \omega^2$$

$$= \frac{1}{2} M \left(\frac{1}{2} \omega \right)^2 + \frac{1}{2} \frac{1}{12} ml^2 \omega^2$$

$$\omega = \sqrt{\frac{3g(1 - \cos \theta)}{l}}$$

Now horizontal component $v_H = v \cos \theta$

$$= \omega \frac{1}{2} \cos \theta$$

$$v_H = \sqrt{\frac{3gl}{4} (1 - \cos \theta)} \cos \theta$$

$$\frac{dv_H}{d\theta} = 0 \text{ gives } \cos \theta = \frac{2}{3} \Rightarrow v_H = \frac{\sqrt{gl}}{3}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (B)

2. Ans. (D)

3. Ans. (B)

Sol. Na_2CO_3 does not react with acidic KMnO_4 solution.

4. Ans. (B)

Sol. Due to stable electronic configuration.

5. Ans. (C)

6. Ans. (D)

7. Ans. (A,C,D)

8. Ans. (A,B)

9. Ans. (A,B,C,D)

10. Ans. (B,D)

Sol. Cd^{2+} is soluble in excess of $\text{NH}_3(\text{aq.})$

KNO_3 on heating does not liberate NO_2 (g)

11. Ans. (A,B,C,D)

Sol. (A) $\text{MnO}_4^{2-} + \text{H}^+ \longrightarrow \text{MnO}_4^- + \text{MnO}_2$

(B) $\text{Cl}_2 + \text{NaOH} (\text{hot \& conc.}) \longrightarrow \text{NaCl} + \text{NaClO}_3$

(C) $\text{Cu}^+ \xrightarrow[\text{Solution}]{\text{in aqueous}} \text{Cu}^{2+} + \text{Cu}$

(D) $\text{H}_3\text{PO}_3 \xrightarrow{\Delta} \text{PH}_3 + \text{H}_3\text{PO}_4$

12. Ans. (B,C,D or C,D)

13. Ans. (A, B, C, D)

14. Ans. (A,B,C)

15. Ans. (D)

16. Ans. (B)

17. Ans. (A)

18. Ans. (B)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (D)**

Sol. The required arrangements is

$$0, \pm 1, \pm 2, \dots, \pm 10$$

So, required number

$$= 1 \times (2 \times 2 \times \dots \text{upto } 10 \text{ factors})$$

$$= 2^{10}$$

2. **Ans. (A)**

Sol. $\alpha + \beta > \gamma$, $\beta + \gamma > \alpha$, $\gamma + \alpha > \beta$

$$\Rightarrow \frac{\gamma}{\beta + \alpha} < 1, \frac{\alpha}{\beta + \gamma} < 1, \frac{\beta}{\gamma + \alpha} < 1$$

$$\Rightarrow \frac{\alpha}{\beta + \gamma} + \frac{\beta}{\gamma + \alpha} + \frac{\gamma}{\alpha + \beta} < 3$$

$$\text{Also, we hence, } (\alpha + \beta + \gamma) \left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \right) \geq 9$$

$$\Rightarrow \frac{\alpha + \beta + \gamma}{\alpha} + \frac{\alpha + \beta + \gamma}{\beta} + \frac{\alpha + \beta + \gamma}{\gamma} \geq 9$$

$$\Rightarrow \frac{\alpha + \beta}{\gamma} + \frac{\beta + \gamma}{\alpha} + \frac{\gamma + \alpha}{\beta} \geq 6 \quad \dots(1)$$

using A.M. $\geq$ H.M.

$$\Rightarrow \frac{\alpha}{\beta + \gamma} + \frac{\beta}{\gamma + \alpha} + \frac{\gamma}{\alpha + \beta} \geq \frac{9}{\frac{\beta + \gamma}{\alpha} + \frac{\gamma + \alpha}{\beta} + \frac{\alpha + \beta}{\gamma}} \quad \dots(2)$$

from (1) and (2)

$$\Rightarrow \frac{\alpha}{\beta + \gamma} + \frac{\beta}{\gamma + \alpha} + \frac{\gamma}{\alpha + \beta} \geq \frac{3}{6} = \frac{3}{2}$$

$$\Rightarrow \left[\frac{\alpha}{\beta + \gamma} + \frac{\beta}{\gamma + \alpha} + \frac{\gamma}{\alpha + \beta} + \frac{1}{2} \right] \neq 1, 4, 5$$

3. **Ans. (B)**

$$\text{Sol. } I_k = \int_{-k\pi}^{k\pi} |\sin x| [\sin x] dx \quad \dots(1)$$

$$\text{using } \int_a^b f(x) dx = \int_a^b f(a + b - x) dx$$

$$I_k = \int_{-k\pi}^{k\pi} |\sin x| [-\sin x] dx \quad \dots(2)$$

(1) + (2)

$$\Rightarrow 2I_k = \int_{-k\pi}^{k\pi} |\sin x| ([\sin x] + [-\sin x]) dx$$

$$= 2 \int_0^{k\pi} |\sin x| (-1) dx = (-2) k \times 2 = -4k$$

$$\Rightarrow I_k = -2k$$

$$\text{Now } \frac{1}{I_k I_{k+1}} = \frac{1}{(-2k)(-2(k+1))}$$

$$\text{so, } \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{I_k I_{k+1}} = \frac{1}{4}$$

4. **Ans. (B)**

Sol. A is an orthogonal matrix

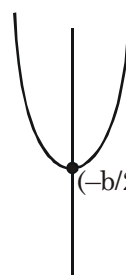
$$\Rightarrow a^2 + b^2 + c^2 = 1, ac + ba + cb$$

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$$

$$= 1 + 0 = 1$$

$$\Rightarrow |a + b + c| = 1$$

5. **Ans. (A)**



Sol.

$$-\frac{b}{2a} = 4 \Rightarrow b = -8a$$

$$-\frac{(b^2 - 4ac)}{4a} = 2 \Rightarrow b^2 - 4ac = -8a$$

$$\Rightarrow 64a^2 - 4ac = -8a \Rightarrow 64a - 4c = -8$$

$$c = \frac{64a + 8}{4} = 16a + 2$$

$$\text{Now } (abc) = a(-8a) 2(8a + 1)$$

$$= (-16)(8a^3 + a^2)$$

$$\text{minimum of } (abc) = (-16)(8 \times 4^3 + 4^2)$$

$$\text{maximum of } (abc) = (-16)(8 \times 2^3 + 2^2)$$

$$\text{Ratio of minimum to maximum} = \frac{132}{17}$$

6. **Ans. (C)**

Sol. put $x = -x$

$$\Rightarrow \text{we get } f(x) = -f(-x)$$

$$\Rightarrow f'(x) = +f'(-x)$$

$$\Rightarrow f''(x) = -f''(-x)$$

$$\Rightarrow f^{2020}(0) = -f^{2020}(0)$$

$$\Rightarrow f^{2020}(0)$$

7. **Ans. (A,D)**

Sol. As $\vec{a}, \vec{b}$ are non collinear unit vectors

$\Rightarrow \vec{a}, \vec{b}, \vec{a} \times \vec{b}$ are non-coplanar non-zero vectors

$$\Rightarrow a - b = b - c = c - a = 0$$

8. Ans. (B,D)

Sol. Let (h,k) be the point of contact of tangent drawn from (1,2)

$$\Rightarrow \text{slope of the tangent} = \frac{k-2}{h-1}$$

$$\Rightarrow \frac{dy}{dx} \Big|_{\text{at}(h,k)} = \frac{k-2}{h-1}$$

$$\Rightarrow h^3 - 3h^2 + 4 = 0$$

$$\Rightarrow h = -1, 2$$

$$\text{At } h = -1, (k-2)^2 = -6$$

$$\Rightarrow \text{no real } k$$

$$\text{At } h = 2, (k-2)^2 = 12$$

$$\Rightarrow k = 2 + \sqrt{12}, 2 - \sqrt{12}$$

There can be two tangent from (1,2)

equation of tangent at $(2, 2+2\sqrt{3})$ is

$$y - (2+2\sqrt{3}) = 2\sqrt{3}(x-2)$$

9. Ans. (B,C)

Sol. $x + y + z = 10, x, y, z \in \mathbb{N}$

Put $x = a + 1, y = b + 1, z = c + 1, a, b, c \in \mathbb{N}$

$$\Rightarrow a + b + c = 7$$

Total number of solutions

$$= {}^{7+3+1}C_7 = \frac{9 \times 8}{2} = 36$$

Number of solutions when c is odd = 16

$$\text{probability that } z \text{ is even} = \frac{16}{36} = \frac{4}{9}$$

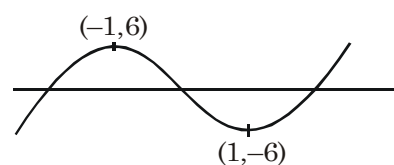
$$\text{probability that } z \text{ is odd} = \frac{20}{36} = \frac{5}{9}$$

10. Ans. (B,D)

Sol. $f'(x) = 7x^6 - 7$

sign scheme for $f'(x)$

graph of $y = x^2 - 7x$



$f(x) = 0$ has only one real root

$$\Rightarrow -a > 6 \text{ or } -a < -6$$

$$\Rightarrow a < -6 \text{ or } a > 6$$

$f(x) = 0$ has three real distinct roots

$$\Rightarrow -6 < a < 6$$

11. Ans. (B,C)

$$\text{Sol. } \frac{dy}{dx} + \frac{2xy}{x^2-1} = \frac{2x}{(x^2-1)^2}$$

$$\text{I.F.} = e^{\int \frac{2x}{(x^2-1)^2} dx} = e^{\ln(x^2-1)} = (x^2-1)$$

so, solution is given by

$$y \times (x^2-1) = \int \frac{2x}{(x^2-1)^2} (x^2-1) dx$$

$$\Rightarrow y \times (x^2-1) = \ln|(x^2-1)| + C$$

$$\text{put } x = \sqrt{2}, y = 0$$

$$\text{we get } c = 0$$

$$\Rightarrow y = \frac{\ln(x^2-1)}{(x^2-1)}, x > 1$$

$$\frac{dy}{dx} = \frac{1-2x\ln(x^2-1)}{(x^2-1)^2}$$

Both positive and negative

12. Ans. (C,D)

Sol. d = common difference

$$\frac{S_m}{S_n} = \frac{\frac{m}{2} [2a + (m-1)d]}{\frac{n}{2} [2a + (n-1)d]}$$

$$= \frac{m(10 + (m-1)d)}{n(10 + (n-1)d)}$$

$$= \frac{5((10-\lambda) + md)}{(10-d) + nd}$$

$$\frac{S_m}{S_n} \text{ is independent of } n$$

$$\Rightarrow d = 0, 10$$

$$a_2 = 5, 15$$

13. Ans. (A,B)

Sol. Let A and B be two square matrices of order 3

$$\text{If } A = B^2$$

$$\Rightarrow |A| \text{ can't be negative}$$

14. Ans. (B,C,D)

Sol. Let the variable chord be AB where

$$\text{equation is } x \cos \alpha + y \sin \alpha = p$$

Now joint equation of OA and OB is

$$\left(\frac{x^2}{9} - \frac{y^2}{b^2} \right) = \left(\frac{x \cos \alpha + y \sin \alpha}{p} \right)^2$$

$$\Rightarrow p^2 = \frac{9b^2}{b^2-9}$$

$\Rightarrow$ equation of the fixed circle is

$$x^2 + y^2 = \frac{9b^2}{b^2-9}$$

for $b = 2$, $R = \sqrt{\frac{9 \times 4}{(-5)}}$ not possible

$$b = 5, R = \sqrt{\frac{9 \times 25}{16}} = \frac{3 \times 5}{4}$$

$$b = 4, R = \sqrt{\frac{9 \times 16}{7}}$$

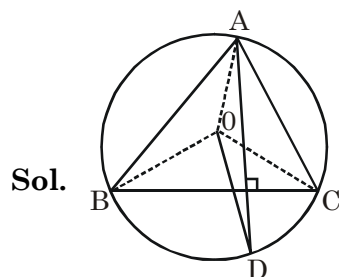
$$(\sqrt{3})^2 + \left(\sqrt{\frac{123}{7}}\right)^2 = 3 + \frac{123}{7}$$

$$= \frac{21+123}{7} = \frac{144}{7} = R^2$$

$$\Rightarrow \left(\sqrt{3}, \sqrt{\frac{123}{7}}\right) \text{ lies on the circle}$$

Solution for Question 15 and 16

15. Ans. (A)



Equation of circumcircle is

$$x^2 + y^2 = 25$$

$$\angle BOD = 2(90 - B) = \pi - 2B$$

$$\angle DOC = 2(90 - C) = \pi - 2C$$

$$\angle AOB = 2C$$

$$\angle AOC = 2B$$

using rotation concept

$$Z_D = Z_B e^{(\pi - 2B)i}$$

Also we have

$$\Rightarrow Z_D = -\frac{Z_B Z_C}{Z_A}$$

$$\text{so, } Z_1 Z_2 Z_3 = -\frac{Z_B Z_C}{Z_A} \frac{Z_C Z_A}{Z_B} \frac{Z_A Z_B}{Z_C}$$

$$= -Z_A Z_B Z_C$$

$$= (-1)(3 + 4i)(-4 + 3i)(4 + 3i)$$

$$= 75 + 100i$$

16. Ans. (B)

Sol. Circumcenter (0,0) centroid $\left(1, \frac{10}{3}\right)$

$$\Rightarrow \text{orthocenter } (3,10)$$

Solution for Question 17 and 18

Sol. Point of intersection of $y^2 = (e^{-kr})x$ and

$$y = mx \text{ are } (0,0) \text{ and } \left(\frac{e^{-kr}}{m^2}, \frac{me^{-kr}}{m^2}\right)$$

$$\text{So } A_r = \int_0^{\frac{e^{-kr}}{m^2}} \left(\sqrt{e^{-kr}} \sqrt{x} - mx\right) dx$$

$$= \frac{e^{-2kr}}{6m^3}$$

$$\Rightarrow \frac{A_{r+1}}{A_r} = e^{-2k} = \text{constant}$$

$$\Rightarrow A_1, A_2, A_3 \dots \text{ are in G.P.}$$

$$\text{Now } \lim_{n \rightarrow \infty} \sum_{r=1}^n A_r = \lim_{n \rightarrow \infty} \frac{1}{6m^3} \sum_{r=1}^n e^{-2kr}$$

$$= \frac{1}{6m^3 (e^{2k} - 1)}$$

$$\Rightarrow \frac{1}{6m^3 (e^{2k} - 1)} = \frac{4}{3(e^{2k} - 1)}$$

$$\Rightarrow m = \frac{1}{2}$$

17. Ans. (B)

18. Ans. (A)